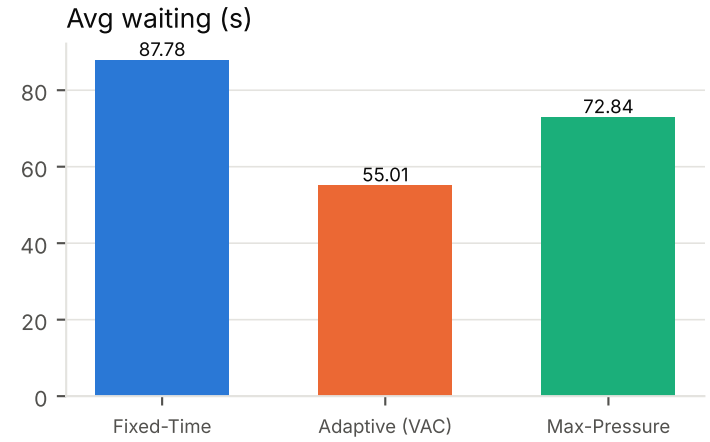
VAC-Inspired Adaptive Traffic Control System - Experiment Report

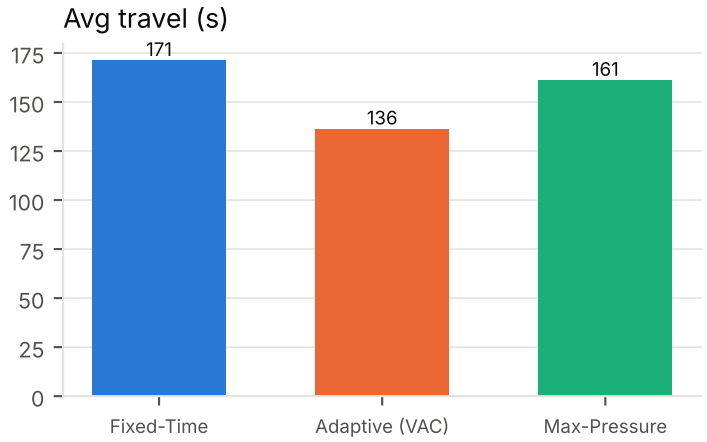
Generated 2026-11-10 07:37 | 9 experiment(s) | all values measured in SUMO via TraCI

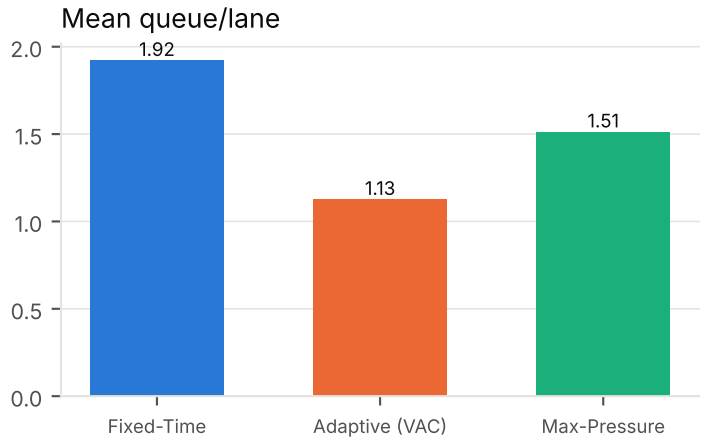
Scenario(s): high Seed(s): 1, 2, 3 Duration(s): 1800 s

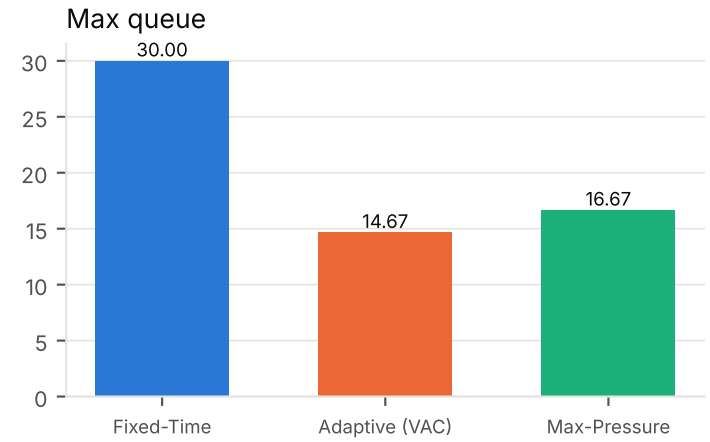
ID	Controller	Scenario	Seed	Avg waiting (s)	Avg travel (s)	Mean queue/lane	Max queue	Throughput	Stops/veh	Phase changes	Latency (ms)	Starvation
19	Fixed-Time	high	1	93.44	177.88	2.03	30	1529	1.97	272	0.0034	0
20	Adaptive (VAC)	high	1	55.60	137.20	1.12	17	1640	1.68	311	0.0064	0
21	Max-Pressure	high	1	76.87	165.45	1.59	17	1574	2.90	527	0.0041	6
22	Fixed-Time	high	2	89.37	173.04	2.05	33	1580	1.96	272	0.0031	0
23	Adaptive (VAC)	high	2	57.95	139.52	1.22	14	1686	1.70	303	0.0063	0
24	Max-Pressure	high	2	70.05	158.33	1.50	18	1643	2.89	537	0.0041	7
25	Fixed-Time	high	3	80.53	162.63	1.68	27	1597	1.82	272	0.0030	0
26	Adaptive (VAC)	high	3	51.48	132.12	1.04	13	1653	1.67	321	0.0064	0
27	Max-Pressure	high	3	71.59	159.43	1.45	15	1613	2.86	528	0.0041	10

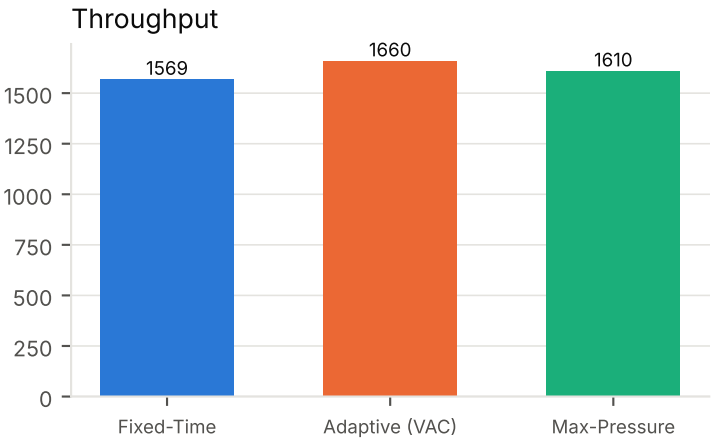
Controller comparison (mean over selected experiments)

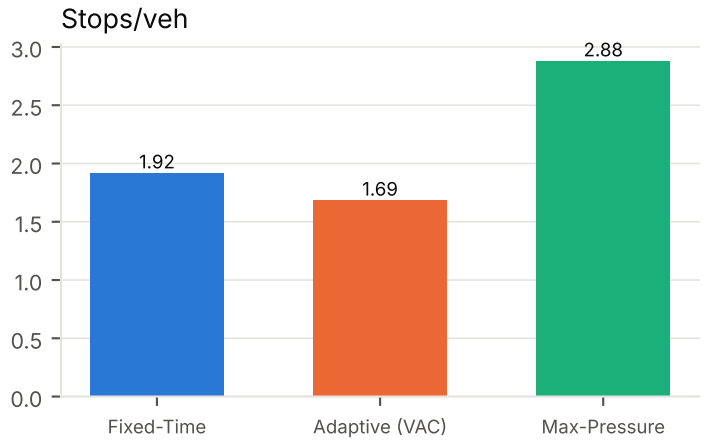


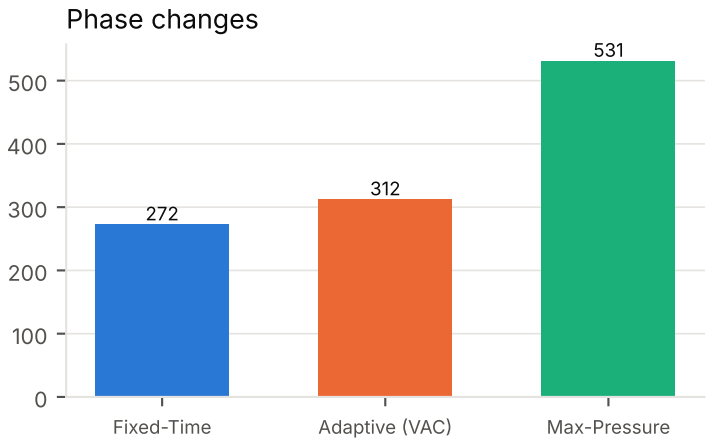


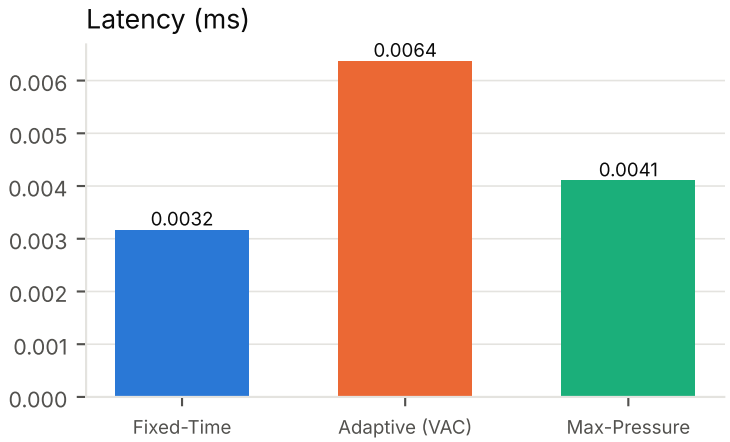


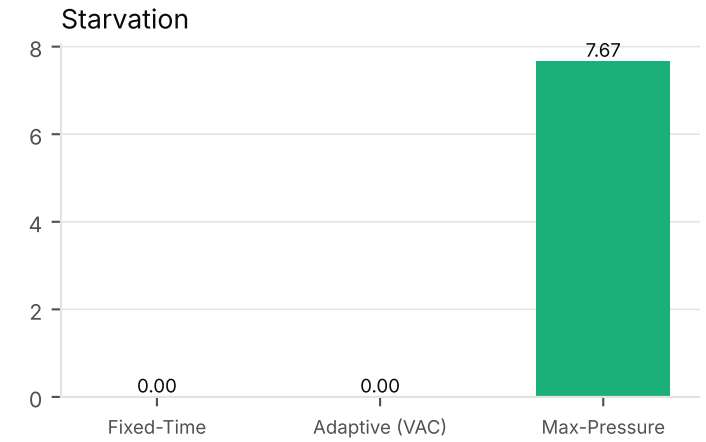












Time series (one line per experiment)

